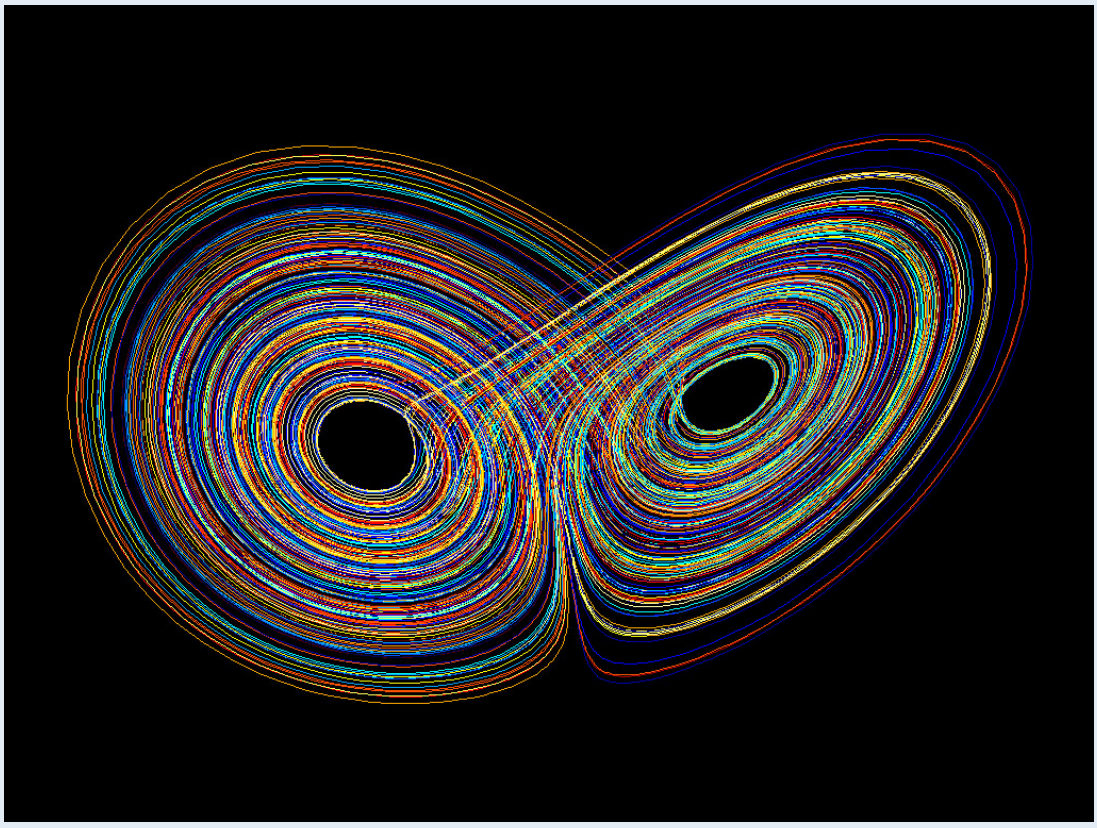
ARythm

Chaos to Control

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Contents

1 Introduction **2**

2 Logistic Map **2**

3 Linear Systems and State Space **4**

4 Bifurcations **5**

5 Limit Cycles **7**

6 Control Design **8**

List of Figures

1 Logist Map time-series example 3
2 Bifurcation diagram of the Logistic Map 4
3 A 6
4 B 6
5 C 6
6 D 6
7 Stable Limit Cycle between Susceptible and Infected Populations 7

1 Introduction

Dynamical Systems and Control Theory form the backbone of modern mathematical modeling, helping us understand, predict and control everything from population growth, epidemics and climate patterns to combustion engines, financial markets and social networks.

The reason this doc is long is because it contains a lot of pictures, simple theory, resources and intuition. So do not procrastinate or get intimidated. The questions are short and simple.

More importantly, please contact us for any hints and doubts as this is not a contest or an assignment but an educational and fun task where we are going to explore fundamental concepts in dynamical systems and control theory!

2 Logistic Map

Context

Please watch Veritasium's video on the logistic map for all the intuition you need!

"In studying the logistic map, we see a perfect example of how simple nonlinear systems can exhibit a vast complexity of behaviors, from stable fixed points to periodic oscillations and chaotic regimes."

- Robert M. May, a pioneer in the study of chaos in ecological systems

Real-World Application

The logistic map is used in modeling fish populations in confined environments, predicting market price fluctuations, and understanding neural firing patterns in the brain.

Numerical Experimentation

Let the evolution of population from one generation to another be given by: $f(x) = rx(1-x)$
Tasks:

(a) Implement $f(x)$ in Python

(b) Create a function to calculate and store the sequence $x_{n+1} = rx_n(1-x_n)$

(c) Plot the sequence $\{x_n\}$ against n for the experiment given below

Experiment: Using $x_0 \in (0,1)$, analyze the behavior (mention number of limiting values of x) for:

1. $r = 0.5$ → 1
2. $r = 2.0$ → 1
3. $r = 3.3$ → 2
4. $r = 3.46$ → 64
5. $r = 3.56$ → 9

$x_0 = 0.01$
$n = 500$
excluded / transient = 100

Your output should be a time-series like:

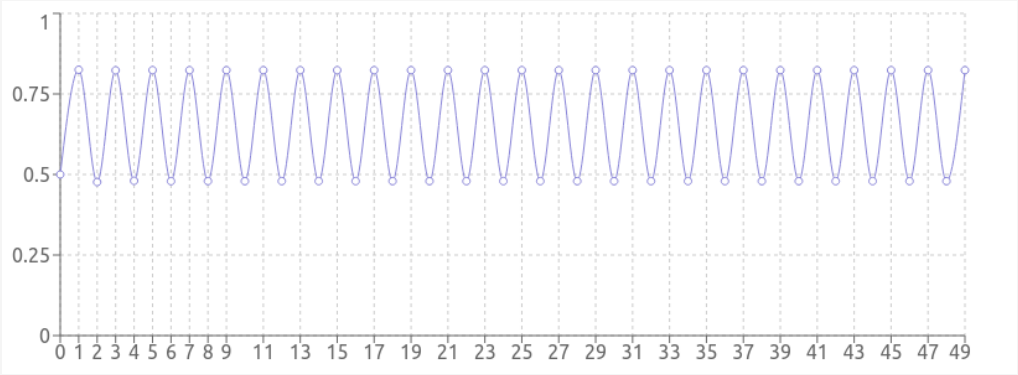


Figure 1: Logist Map time-series example

Mathematical Analysis

From the bifurcation diagram of the logistic map analyse:

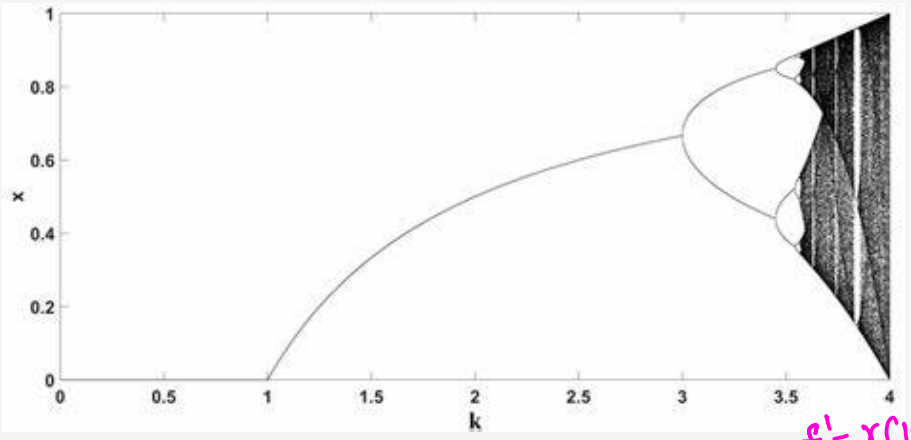


Figure 2: Bifurcation diagram of the Logistic Map

- Fixed points of the logistic map and their stability
- Find the value of r (closed-form) for the first, second and third bifurcation points

  (Note : Solve this question after solving the Bifurcations section of this mini-task)

3 Linear Systems and State Space

Context

State Space representation is fundamental concept in modern control theory, providing a powerful framework for analyzing and designing control systems. You may have come across the state space representation (velocity vs displacement) of a spring-mass system in PH1010 or JEE.

Do watch 3B1B's video on matrix exponentiation from time 22:20 for a geometric explanation of a linear system (although you may not use it).

$$\begin{cases} \dot{x} = ax + by \\ \dot{y} = cx + dy \end{cases}$$

(a) Express in matrix form

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\dot{\vec{x}} = A\vec{x}$$

(b) Find the general solution in terms of eigenvalues (λ_1, λ_2) and eigenvectors $(\vec{v}_1, \vec{v}_2)$

$|A - \lambda I| = 0 \Rightarrow (a-\lambda)(d-\lambda) = bc \longrightarrow \lambda_1, \lambda_2$

$(A - \lambda I)\vec{v} = 0 \rightarrow v_1, v_2$

$\vec{x}(t) = c_1 e^{\lambda_1 t}\vec{v}_1 + c_2 e^{\lambda_2 t}\vec{v}_2$ (c_1, c_2 depend on given conditions)

(c) Analyze stability conditions with respect to eigenvalues

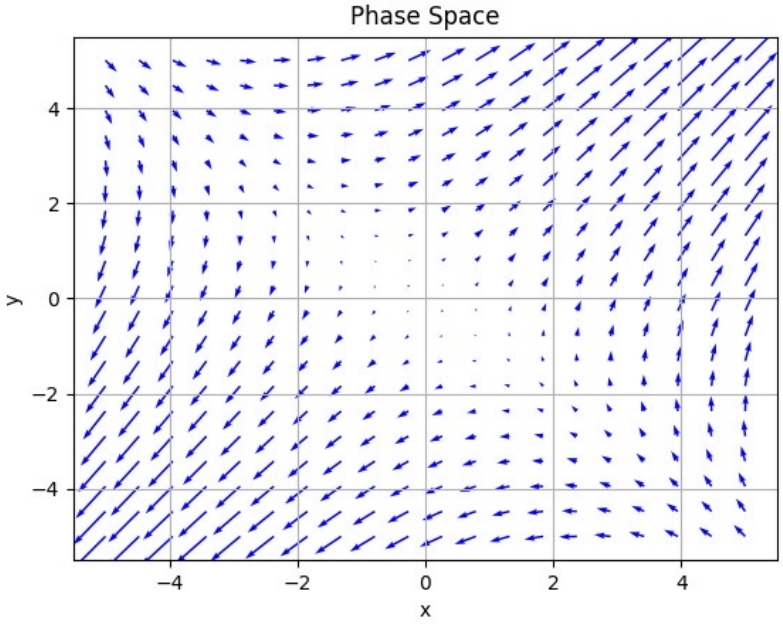
Stable: $\mathrm{Re}(\lambda_1), \mathrm{Re}(\lambda_2) < 0$

unstable: either $\mathrm{Re}(\lambda_1)$ or $\mathrm{Re}(\lambda_2)$ or both > 0

Saddle: if $\lambda_1 > 0$ & $\lambda_2 < 0$ or $\lambda_1 < 0$ & $\lambda_2 > 0$

oscillatory: $\mathrm{Re}(\lambda_1), \mathrm{Re}(\lambda_2) = 0$

(d) Draw by hand/Plot using program the trajectories/phase space for any matrix of you choice.



System Analysis

Consider the system:

$$\begin{cases} \dot{x} = ax + by \\ \dot{y} = cx + dy \end{cases}$$

Tasks:

(a) Express in matrix form

(b) Find the general solution in terms of eigenvalues (λ_1, λ_2) and eigenvectors $(\vec{v}_1, \vec{v}_2)$

(c) Analyze stability conditions with respect to eigenvalues

(d) Draw by hand/Plot using program the trajectories/phase space for any matrix of you choice.

4 Bifurcations

Context

Bifurcations represent qualitative changes in system behavior as parameters (r in the logistic map) vary. They are crucial in understanding critical system transitions and tipping points.

For example in the logistic map at r=3 the graph suddenly splits into two (into a period-2 cycle from a single fixed point)

Real-World Application

Bifurcations appear in:

- Phase transitions in materials
- Sudden onset of diseases in populations
- A Yes or No can push your life into two completely different directions

Analysis Tasks

1. Mention very briefly about Pitchfork, Transcritical, Hopf, and Saddle-Node bifurcations.
2. Do not forget to solve the bifurcation question in the Logistic Map section!
3. Analyse the following 1D system for bifurcations (μ is a parameter of the system which cauese bifurcation)

$$\dot{x} = x^3 - 5x^2 - (\mu - 8)x + \mu - 4$$

1. Mention very briefly about Pitchfork, Transcritical, Hopf, and Saddle-Node bifurcations.

(Used ChatGPT AI)

1. Pitchfork Bifurcation: A bifurcation where a single equilibrium point splits into multiple equilibrium points (usually two or more). Involves symmetry, where the bifurcation creates symmetrical branches of stable and unstable fixed points.
Example: Observed in systems with a symmetry property, such as population dynamics or biological systems.
2. Transcritical Bifurcation: A bifurcation where two fixed points (equilibria) exchange stability. At the bifurcation point, one equilibrium becomes unstable while the other becomes stable, and vice versa as parameters change.
Example: Found in systems like fluid dynamics or chemical reactions where the stability of solutions reverses.
3. Hopf Bifurcation: Occurs when a fixed point becomes unstable, and a limit cycle (oscillatory behavior) emerges. As a control parameter crosses a critical value, complex conjugate pairs of eigenvalues cross the imaginary axis, leading to oscillations.
Example: Seen in systems like circadian rhythms or electrical circuits exhibiting oscillatory behavior.
4. Saddle-Node Bifurcation: A bifurcation where two equilibrium points merge into one or split into two. Involves two fixed points moving towards or away from each other; one becomes unstable and the other stable.
Example: Observed in chemical reactions, mechanical systems, or neural networks where states change due to critical parameter changes.

3. Analyse the following 1D system for bifurcations (μ is a parameter of the system which cauese bifurcation)

$$\dot{x} = x^3 - 5x^2 - (\mu - 8)x + \mu - 4$$

Fixed points: $1,\ 2-\sqrt{\mu},\ 2+\sqrt{\mu}$

$\frac{d\dot{x}}{dx} = 3x^2 - 10x + 8 - \mu$ $\longrightarrow < 0 \rightarrow$ stable

$\longrightarrow > 0 \rightarrow$ unstable

Bifurcation occurs when there is repeated root or

$\frac{d\dot{x}}{dx} = 0$ when $\dot{x} = 0$.

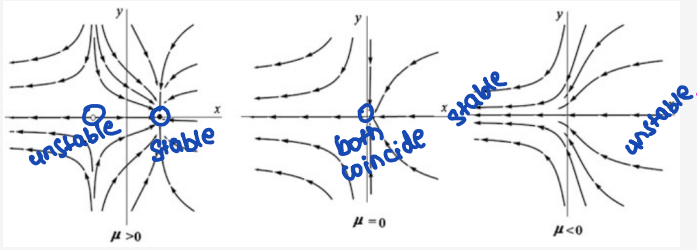
$\mu = 1, 0 \rightarrow$ Bifurcation happens

$\rightarrow$ Saddle-node bifurcation for both

Visual Task

Mention the type of bifurcation in the following 2D systems (Transcritical, Hopf, Pitch fork and Saddle Node) :

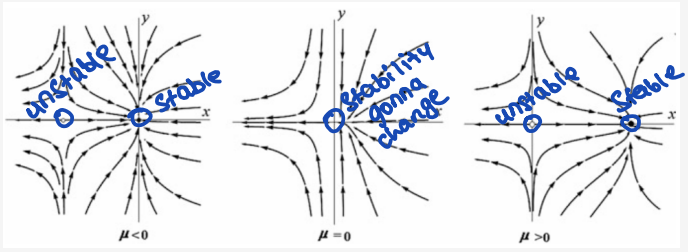
1.



Transcritical

Figure 3: A

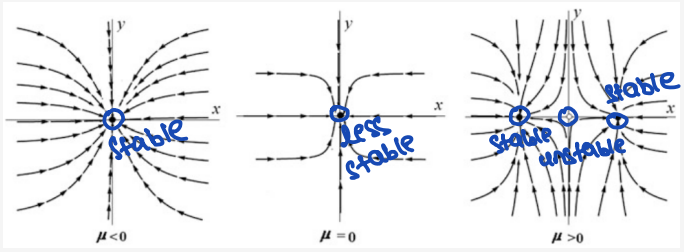
2.



Saddle Node

Figure 4: B

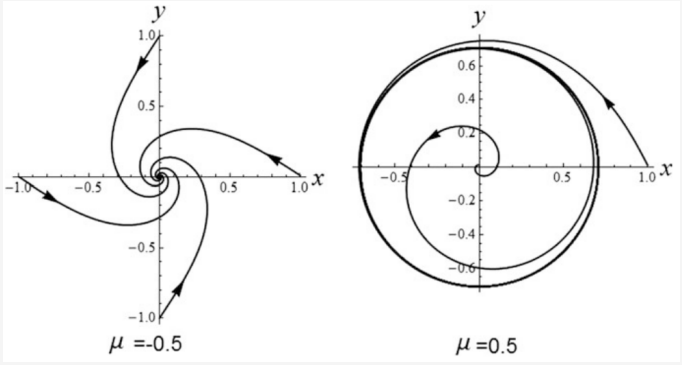
3.



Pitchfork

Figure 5: C

4.



Hopf

Figure 6: D

5 Limit Cycles

Context

Limit cycles represent self-sustained oscillations in nonlinear systems, maintaining their amplitude without external forcing.

Real-World Application

Limit cycles appear in:

- clock signals where we need a constant frequency which it not disturbed by small perterbations
- Heart rhythm dynamics
- Predator-prey populations
- Wind-induced vibrations in structures like airplane wings

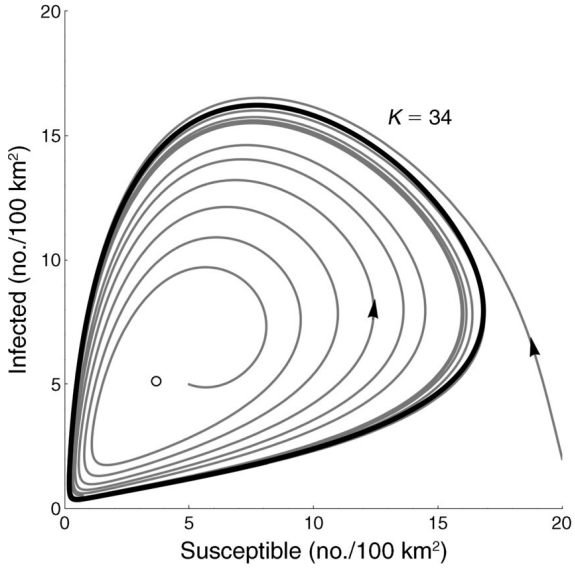


Figure 7: Stable Limit Cycle between Susceptible and Infected Populations

Analysis Problem

1. Show that the system

$$\dot{x} = y + \frac{x}{\sqrt{x^2+y^2}}\left\{1-\left(x^2+y^2\right)\right\}, \dot{y} = -x + \frac{y}{\sqrt{x^2+y^2}}\left\{1-\left(x^2+y^2\right)\right\}$$

   has a stable limit cycle.

2. Are the oscillations of a simple ideal spring-mass system (non-damped) a limit cycle? Justify.

1. Show that the system

$$\dot{x} = y + \frac{x}{\sqrt{x^2+y^2}}\left\{1-(x^2+y^2)\right\}, \dot{y} = -x + \frac{y}{\sqrt{x^2+y^2}}\left\{1-(x^2+y^2)\right\}$$

has a stable limit cycle.

$x = r\cos\theta$, $y = r\sin\theta$

$x\dot{x} + y\dot{y} = r\dot{r} \Rightarrow xy + \frac{x^2}{r}(1-r^2) - xy + \frac{y^2}{r}(1-r^2) = r\dot{r}$

$\Rightarrow \dot{r} = 1-r^2$

$\dot{r}\cos\theta - r\sin\theta\cdot\dot{\theta} = r\sin\theta + \frac{r\cos\theta}{r}\{1-r^2\}$

$\Rightarrow \dot{\theta} = -1$ → constant angular velocity

for larger r, $\dot{r} < 0$ and for smaller r, $\dot{r} > 0$

↓ prevents spirals going to infinity

↓ prevents spirals collapsing into origin

Hence it has a stable limit cycle

2. Are the oscillations of a simple ideal spring-mass system (non-damped) a limit cycle? Justify.

No, the phase space diagrams of spring-mass (undamped) system is concentric ellipses which are neutrally stable. whereas for a limit cycle, other trajectories spiral towards it, which is asymptotically stable.

6 Control Design

Context

State feedback control is a fundamental technique in modern control theory. In order to control the system we give an input u to the system which depends on the state x of the system($u = Kx$). The controlling input is given based on certain constraints. For example, we would not want fluctuations in voltage in our device, therefore we impose a contraint that the state of the system (voltage) should not oscillate.

The eigenvalues of the matrix (in section 3 Sinear Systems and State Space) correspond to the "poles" of the system. We place these poles at desired positions as mentioned above. This is also referred to as Pole Placement Control another very fundamental foundational concept in control theory.

Real-World Application

Applications include:

- Spacecraft attitude control
- Industrial robot manipulation
- Power system stabilization
- Autonomous vehicle control

Design Problem

Given the system $\dot{x} = Ax + Bu$ where:

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The closed-loop system with state feedback is:

$$\dot{x} = (A - BK)x$$

Find the range/set of feedback gains $K = [k_1 \quad k_2]$ such that:

- The closed-loop system is stable
- The system response has no oscillations

Given the system $\dot{x} = Ax + Bu$ where:

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$2-K_1+K_2$
$0 \quad 0$

The closed-loop system with state feedback is:

$$\dot{x} = (A - BK)x$$

Find the range/set of feedback gains $K = [k_1 \quad k_2]$ such that:

- The closed-loop system is stable → eigen values have -ve real parts
- The system response has no oscillations → eigen values must be real and negative

$$\dot{X} = \begin{bmatrix} 2-K_1 & 1-K_2 \\ -1 & 3 \end{bmatrix} X$$

characteristic eqn of closed loop system is

$$(2-K_1-\lambda)(3-\lambda)-(-1)(1-K_2)=0$$

$$\lambda^2-(5-K_1)\lambda+(7-3K_1-K_2)=0$$

$$\Delta = K_1^2-10K_1+25-28+12K_1+4K_2 = K_1^2+2K_1+4K_2-3>0$$

-ve roots : $5-K_1<0 \Rightarrow K_1>5$ $\quad 7-3K_1-K_2>0 \Rightarrow K_2<7-3K_1$

